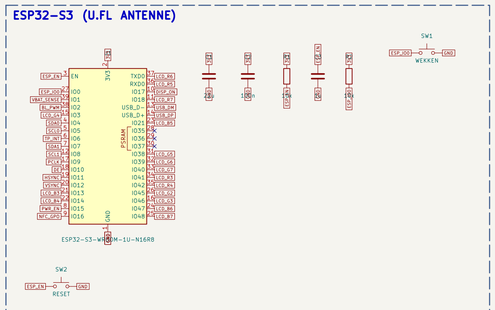
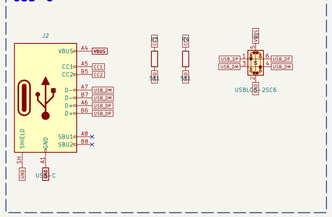
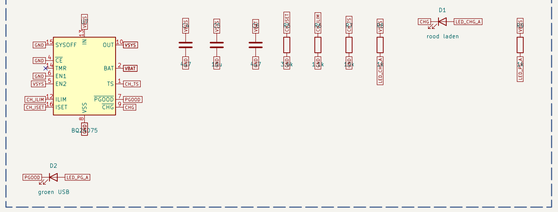
ESP32-S3 (U.FL ANTENNE)



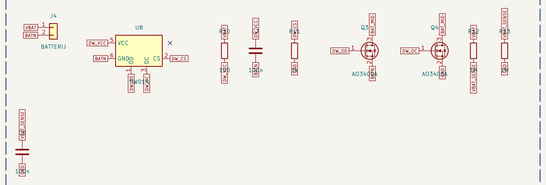
USB-C



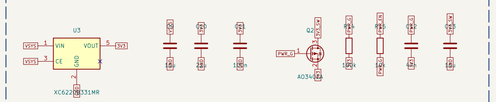
LADEN (BQ24075)



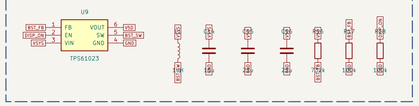
BATTERIJ + BEVEILIGING



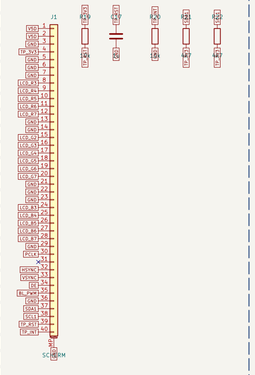
3,3 V + SENSORVOEDING



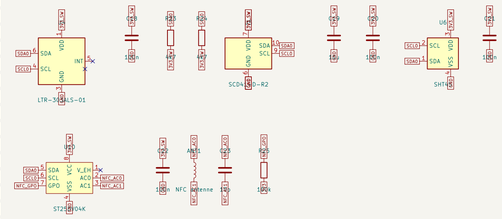
SCHERM 5 V (TPS61023)



SCHERM 4.3" (FFC)



SENSOREN + NFC (I2C0)



BEVESTIGING

M1 M2 M3 M4 M5 M6

LUCHTMETER S3 v2.0 — ESP32-S3-WROOM-1U (external antenna) behind a Waveshare 4.3in 800x480 IPS capacitive touchscreen (generated from tools/design.py)
Screen FFC J1 (40p 0.5 mm, RGB565): physical pin k = screen pin 41−k. R3-7 IO41 IO42 IO44 IO43 IO18, G2-7 IO45 IO46 IO3 IO38 IO39 IO40, B3-7 IO13 IO14 IO21 IO47 IO48, PCLK IO9 DE IO10 HSYNC IO11 VSYNC IO12, backlight PWM IO2.
Touch GT911 on I2C1 (SDA IO7, SCL IO8, INT IO6, address 0x5D). Sensors + NFC on I2C0 (SDA IO4, SCL IO5): SCD41 0x62, SHT45 0x44, LTR-303 0x29, ST25DV 0x53/0x57/0x2D.
Power: USB-C -> BQ24075 (charge 250 mA, power path) -> LiPo on BAT+/BAT- (DW01A) -> XC6220 3.3 V. Screen 5 V from TPS61023 (DISP_ON IO17 high). Sensors + NFC on 3V3_SW (PWR_EN IO15 low = on).
Wake from deep sleep: WEKKEN button (IO0) or NFC field (ST25DV GPO -> IO16). Battery voltage = ADC(IO1) x 2.

ESP32-S3-WROOM-1U + 4.3in touch + SCD41 + SHT45 + LTR-303 + NFC, LiPo
Kobe Van Robaeys — maakleerplek
Sheet: /
File: luchtmeter.kicad_sch
Title: Luchtmeter S3
Size: A1 | Date: 2026-10-18 | **Rev: 1.0**
KiCad E.D.A. 10.0.3 | Id: 1/1